
Optimal-Scaling Analytic-Extension Kernels in an Explicit FCC–QSP LCHS Circuit

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Abstract

1 Linear-combination-of-Hamiltonian-simulations (LCHS) algorithms turn dissipative
2 evolution into a coherent integral of unitary evolutions. Existing explicit,
3 economical circuits use an exact kernel whose cutoff grows as $\log^{1/\beta}(1/\epsilon_{\text{LCHS}})$,
4 whereas recently derived approximate kernels attain the optimal logarithmic de-
5 pendence but had not been tested in that circuit. We insert the published analytic-
6 extension f_2 kernel into the held-fixed sinusoidal Fejér–Clenshaw–Curtis (FCC)
7 quadrature and single-QSP selector, and recompile the kernel-dependent weight
8 preparation with QSVT. Sixteen complete circuits, including explicit advection–
9 diffusion block encodings, are executed with the QUCF state-vector emulator.
10 Across six predeclared contour targets from $10^{-1.5}$ to 10^{-4} , a fit $Q_A =$
11 $a[\log(1/\epsilon_{\text{LCHS}})]^p$ gives $p = 1.399 \pm 0.093$ for the exact $\beta = 0.7$ kernel and
12 $p = 0.708 \pm 0.192$ for the analytic extension; the exponent difference is significant
13 under the predeclared residual-based test ($z = 3.24$, two-sided $p = 1.19 \times 10^{-3}$)
14 and both confidence intervals contain their theoretical values $1/\beta$ and 1. At
15 $(n_x, n_k, t, \epsilon_{\text{LCHS}}) = (6, 6, 0.8, 10^{-4})$, the substituted circuit uses 127,728 rather
16 than 159,660 explicit A -oracle queries (20.0% fewer), while improving the phase-
17 aligned final-state error from 0.4917 to 0.4604 and the all-zero-ancilla probability
18 from 0.209 to 0.691. A resolution sweep locates the practical crossover: fixed
19 amplification overhead makes the exact kernel cheaper at the coarse target, while
20 the analytic extension dominates all three reported outcomes once its smaller cut-
21 off takes over. These results close the mathematical-kernel/explicit-circuit gap
22 without changing the selector, quadrature, qubitization, or PDE block encoding.

23 1 Introduction

24 For $A = L + iH$, with $L = (A + A^\dagger)/2 \succeq 0$ and $H = (A - A^\dagger)/(2i)$, LCHS represents the
25 nonunitary propagator by a coherent superposition of Hamiltonian evolutions,

$$e^{-At} = \int_{\mathbb{R}} g(k) e^{-i(kL+H)t} dk. \quad (1)$$

26 This construction combines optimal initial-state access with standard Hamiltonian-simulation prim-
27 itives [1, 2]. Its precision cost, however, is controlled by the Fourier kernel. Exact LCHS ker-
28 nels cannot attain a cutoff linear in $\log(1/\epsilon_{\text{LCHS}})$ under the standard analyticity assumptions [2].
29 The widely useful β family therefore has the near-optimal radius $K = \mathcal{O}(\log^{1/\beta}(1/\epsilon_{\text{LCHS}}))$. The
30 missing logarithmic behavior is the precision benchmark set by optimal linear-differential-equation
31 solvers [4].

32 Approximate LCHS evades that no-go theorem by replacing exact exponential decay with a certi-
33 fied analytic extension. Low and Somma introduced a rational–Gaussian f_2 family with bounded

LCU normalization and $R = \mathcal{O}(\log(1/\epsilon_{\text{LCHS}}))$ [16]; independently, Schrödingerization work identified smooth auxiliary functions with optimal matrix-query dependence [12]. These mathematical advances do not by themselves establish their performance in the particularly compact explicit implementation of Novikau and Joseph [18]: a sinusoidal FCC grid, one QSP Hamiltonian selector, and QSVT-compatible weight preparation. That separation is scientifically consequential. At finite n_k , a faster-decaying continuous kernel can still lose to amplification, weight synthesis, or quadrature resolution.

We report an end-to-end integration and digital emulation of the published f_2 analytic extension in this FCC-QSP architecture. The contribution is not a new f_2 formula. It is an explicit, auditable kernel-selectable circuit and a matched experiment that isolates the mathematical substitution. Specifically:

- We optimize and certify the published contour bound before any PDE emulation, then map the selected radius to the existing selector’s segment convention.
- We synthesize each complex square-root FCC weight as an even QSVT magnitude polynomial and an exact basis-controlled phase table. All sixteen weight circuits are independently emulated before use.
- On six precision targets, the measured expanded A -oracle count has a lower fitted logarithmic exponent, while the headline state accuracy and success probability both improve. Three cross-resolution experiments expose the finite-precision crossover instead of suppressing it.

2 Related work

The original LCHS identity and its optimal state-preparation cost were developed by An et al. [1]; the exact β -kernel family and near-optimal dependence on all parameters followed in An et al. [2]. Schrödingerization is a closely related dilation viewpoint [9, 11]. Novikau and Joseph [18] supplied the implementation used here: the sinusoidal change of variables makes the quadrature FCC-equivalent and lets one QSP circuit select all Hamiltonian evolutions. Their earlier advection-diffusion and Koopman-von Neumann construction motivates the benchmark [17].

Optimal approximate LCHS has two directly relevant lines. The analytic-extension construction of Low and Somma [16] proves the f_2 formula, contour error bound, bounded normalization, and logarithmic radius used below. It also proposes a multiplexed selector; we deliberately do not use that selector. The smooth Schrödingerization initializations of Jin et al. [12] reach optimal matrix queries through a different implementation. Earlier Fourier-based QSP already used analytic approximations to exponential decay [7]; exponentially convergent quadrature for analytic integrands supplies the associated classical approximation principle [19]. QSP, qubitization, QSVT, LCU, and amplitude amplification provide the underlying primitives [5, 6, 8, 14, 15]. After these kernel results, amplitude-phase separation placed LCHS in a broader framework and derived optimal or fast-forwardable alternatives [10]; changing to that architecture is outside the controlled comparison here. New analytic angle formulae based on orthogonal polynomials [3] are complementary to, but not used by, our numerical angle synthesis.

3 Analytic-extension LCHS in the fixed circuit

3.1 Kernels and certified parameter selection

We use the Fourier convention in Eq. (1). The exact reference is the established $\beta = 0.7$ kernel

$$g_\beta(k) = \frac{1}{2\pi e^{-2\beta} \exp[(1+ik)^\beta](1-ik)}.$$
 (2)

Its circuit cutoff K is the smallest positive multiple of four for which a 50-decimal evaluation of $\int_{|k|>K} |g_\beta(k)| dk$ is at most ϵ_{LCHS} .

The substituted kernel is the published $j = 2, y = 1$ analytic extension [16], divided by $\sqrt{2\pi}$ to match our convention:

$$g_2(k; \gamma, c) = \frac{e^c \exp[-(k^2 + 1)/(4\gamma^2)] e^{-ick}}{\pi (1 + k^2)}.$$
 (3)

Table 1: Pre-validation cutoffs and certified analytic-extension parameters. K_β is the exact-kernel circuit cutoff; K_2 is the rounded analytic-extension cutoff. Every row is generated from the 50-decimal certificate log.

ϵ_{LCHS}	K_β	c	γ	R	y_0	α_R	K_2
3.2×10^{-2}	16	0.742	1.846	4.191	6.887	1.457	8
1.0×10^{-2}	20	0.832	1.996	5.642	8.396	1.656	8
3.2×10^{-3}	28	0.889	2.155	7.166	10.002	1.805	8
1.0×10^{-3}	40	0.928	2.314	8.754	11.674	1.921	12
3.2×10^{-4}	48	0.955	2.470	10.398	13.396	2.014	12
1.0×10^{-4}	60	0.976	2.623	12.090	15.160	2.089	16

Equivalently, in the common $g(k) = \xi(k)/(1 - ik)$ notation, the explicitly compiled numerators are

$$\xi_2(k) = \frac{e^c e^{-(k^2+1)/(4\gamma^2)} e^{-ick}}{\pi(1+ik)}, \quad \xi_\beta(k) = \frac{1}{2\pi e^{-2\beta} e^{(1+ik)^\beta}}.$$

On a lower strip of height $y_0 > 1$, $(z + i)g_2(z)$ is analytic, the pole at $-i$ has the required residue, and g_2 decays uniformly on every fixed strip. The approximate-LCHS theorem therefore gives the operator-norm certificate

$$\delta_{\text{cont}}(R, \gamma, c, y_0) \leq \int_{|k|>R} |g_2(k)| dk + \int_{\mathbb{R}} |g_2(k - iy_0)| dk. \quad (4)$$

For each target, a fixed 36-start constrained SLSQP search minimizes $\alpha_R R$, where $\alpha_R = \int_{-R}^R |g_2(k)| dk$, subject to Eq. (4) $\leq \epsilon_{\text{LCHS}}$. It uses no ADE state or emulator output; the selected solution is recertified by 50-decimal quadrature. Finally, $K = 4\lceil R/4 \rceil$ respects the unchanged selector segmentation. The parameters in Table 1 reproduce the published f_2 cost optima at the shared decade points [16]; our scientific comparison concerns their compilation into a different explicit selector.

Theorem 2 of Low and Somma [16] chooses $\gamma = \Theta(\sqrt{\log(1/\epsilon_{\text{LCHS}})})$ and $R = \Theta(\log(1/\epsilon_{\text{LCHS}}))$ for constant c , while α_R remains bounded. In contrast, Eq. (2) requires $K_\beta = \Theta(\log^{1/\beta}(1/\epsilon_{\text{LCHS}}))$. Because the fixed selector cost is linear in K , the predicted precision exponents are 1 and $1/\beta = 1.429$, respectively.

3.2 FCC weights and the recompiled QSVT path

Both methods use exactly $N_k = 2^{n_k}$ nodes

$$\theta_j = -\frac{\pi}{2} + \frac{j\pi}{N_k - 1}, \quad k_j = K \sin \theta_j, \quad w_j = K \cos \theta_j \frac{\pi}{N_k - 1} g(k_j). \quad (5)$$

Thus only g and its resulting weights differ. The implementation uses a matched pair of complex square-root preparations. The right state has amplitudes proportional to $\sqrt{w_j}$ and the left state to $\sqrt{w_j^*}$; applying the inverse left preparation after the selector therefore gives $(\sqrt{w_j^*})^* \sqrt{w_j} = w_j$, retaining the kernel phase. Let $s_j = \sin \theta_j$ and

$$m_j = 0.9 \frac{\sqrt{|w_j|}}{\max_\ell \sqrt{|w_\ell|}}, \quad r_j = m_j e^{i \arg(w_j)/2}. \quad (6)$$

We fit m_j with an even Chebyshev polynomial in s_j . Degrees are examined in the frozen sequence 16, 24, 32, 40, 48, 56, 62, 80, 96, 112, 126; the first is retained if its FCC-node error is at most $\min(\epsilon_{\text{LCHS}}/50, 10^{-7})$ and its maximum on a 20,001-point guard grid is at most 0.99. The supplied numerical QuCF angle finder then converts the polynomial into QSVT phases. Exactly generated tables of 2^{n_k} basis-controlled R_z rotations supply $\arg(w_j)/2$ and its negative on the two sides; no phase polynomial is assumed. Kernel selection is a generator option, so baseline and variant traverse the same QSVT and two-sided LCU code. Post-processing parses every executed oracle, checks that the tables are exact negatives, and rejects an input without the ordered right-preparation-selector-inverse-left-preparation structure.

Each of the sixteen resulting weight circuits was executed alone. Degrees are 32 at $n_k = 5$, 62 or 80 at $n_k = 6$, and 126 at $n_k = 7$; the observed maximum complex-amplitude error over every case is 1.28×10^{-13} . The full standalone results, including probability checks and expanded STMC counts, are in Appendix B. This check is stronger than evaluating the classical polynomial: it exercises the synthesized angles, SIN signal oracle, phase table, and QuCF basis convention together.

As a separate end-to-end diagnostic, analysis evaluates the finite FCC sum with exact dense Hamiltonian exponentials and compares it with each normalized circuit output. The largest circuit-to-finite-sum discrepancy across all full runs is 0.0343. This diagnostic is not substituted for a result: every reported final error remains the QuCF state compared directly with the runtime-constructed ADE solution.

3.3 Selector, block encoding, and honest query accounting

The selector, qubitization, and ADE block encoding are unchanged from Novikau and Joseph [18]. The supplied selector contains 5,323 QSP angles and is repeated $K/4$ times. Each of its 5,322 angle gaps is one qubitization walk. The supplied QuCF expansion places both U_C and $U_C^\dagger$ in every walk, so one selector invocation contains $2 \times 5,322(K/4)$ calls to the explicit combined block-encoding interface. Each such call contains the controlled A_H and KA_L component circuits; both are present in the gate total. The analytic-extension LCU normalization reduces raw output probability, so a single state-level Grover iterate was fixed before validation. It invokes the entire LCHS selector three times; the baseline invokes it once. Consequently,

$$Q_A = 2 \times 5,322 (K/4) s, \quad s = \begin{cases} 1, & g_\beta, \\ 3, & g_2. \end{cases} \quad (7)$$

Here a query is one actually expanded U_C or $U_C^\dagger$ circuit call; the analysis also records $2Q_A$ component-block-encoding calls. All three selector calls, both reflections, kernel-weight amplification, and controlled phase rotations are present in the STMC total. Each selector call contains the same explicit five-ancilla sparse encoding of H and KL ; no matrix oracle is cost-free. Total ancillary memory is obtained from the executed HDF5 record as total qubits minus n_x , including the n_k Fourier qubits and every state, weight, selector, and block-encoding ancilla.

4 Experimental protocol

Benchmark. We solve the periodic semidiscrete advection–diffusion equation

$$\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi, \quad qquad v = 1, \quad D = 0.01, \quad (8)$$

on $N_x = 2^{n_x}$ points with $\Delta x = 1/(N_x - 1)$, centered periodic first and second differences, and a normalized Gaussian of center 0.5 and width 0.05. All experiments use $n_x = 6$ and $t = 0.8$. At run time, the analysis constructs the same matrix A from $(v, D, \Delta x)$ and computes $e^{-At} \psi(0)$ by dense eigendecomposition; no final vector occurs in circuit-generation code.

The reported error is global-phase invariant,

$$\epsilon_{\text{state}} = \min_{\phi} \left\| \frac{\psi_{\text{QuCF}}}{\|\psi_{\text{QuCF}}\|_2} - e^{i\phi} \frac{e^{-At} \psi(0)}{\|e^{-At} \psi(0)\|_2} \right\|_2, \quad (9)$$

and success probability is the squared norm of the all-zero-ancilla output.

Frozen sweep and fit. The precision experiment uses $(n_x, n_k, t) = (6, 6, 0.8)$ and the six targets $10^{-1.5}, 10^{-2}, 10^{-2.5}, 10^{-3}, 10^{-3.5}, 10^{-4}$. The independent characterization points are $(n_k, \epsilon_{\text{LCHS}}) = (5, 10^{-2}), (6, 10^{-3}), (7, 10^{-4})$. Throughout the experiments, ϵ_{LCHS} denotes the predeclared continuous-kernel certificate budget; the separately reported end-to-end state error is never relabeled as that target. Duplicate configurations share one execution, giving 16 full circuits. The plan, kernel formula, parameter rule, amplification count, sweep, and regression were timestamped and hashed before validation. We fit

$$\log Q_A = \log a + p \log \log(1/\epsilon_{\text{LCHS}}) \quad (10)$$

by ordinary least squares. Standard errors and the exponent-difference test use residual variance with four degrees of freedom per fit. The alternative was predeclared as directional, but we report the more conservative two-sided p -value. Since emulator counts are deterministic, these intervals quantify finite-grid fit uncertainty, not shot noise.

Table 2: Executed end-to-end QuCF results. The first pair is the predeclared headline; the remaining pairs are the complete kernel-family characterization. Ancillae include the Fourier register and all non-state registers. Commas in STMC and Q_A are digit separators.

Method	Test	$(n_x, n_k, t, \epsilon_{\text{LCHS}})$	Final rel. error	STMC	Q_A	Ancillae	Success prob.
$\beta = 0.7$ exact	ADE	$(6, 6, 0.8, 1.0 \times 10^{-4})$	0.4917	7,920,849	159,660	18	0.209
AE- f_2	ADE	$(6, 6, 0.8, 1.0 \times 10^{-4})$	0.4604	6,355,602	127,728	18	0.691
$\beta = 0.7$ exact	ADE	$(6, 5, 0.8, 1.0 \times 10^{-2})$	0.5988	2,585,205	53,220	17	0.289
AE- f_2	ADE	$(6, 5, 0.8, 1.0 \times 10^{-2})$	0.4279	3,109,470	63,864	17	0.883
$\beta = 0.7$ exact	ADE	$(6, 6, 0.8, 1.0 \times 10^{-3})$	0.5826	5,279,830	106,440	18	0.274
AE- f_2	ADE	$(6, 6, 0.8, 1.0 \times 10^{-3})$	0.1679	4,774,953	95,796	18	0.668
$\beta = 0.7$ exact	ADE	$(6, 7, 0.8, 1.0 \times 10^{-4})$	0.2560	8,099,675	159,660	19	0.215
AE- f_2	ADE	$(6, 7, 0.8, 1.0 \times 10^{-4})$	0.1454	6,520,920	127,728	19	0.585

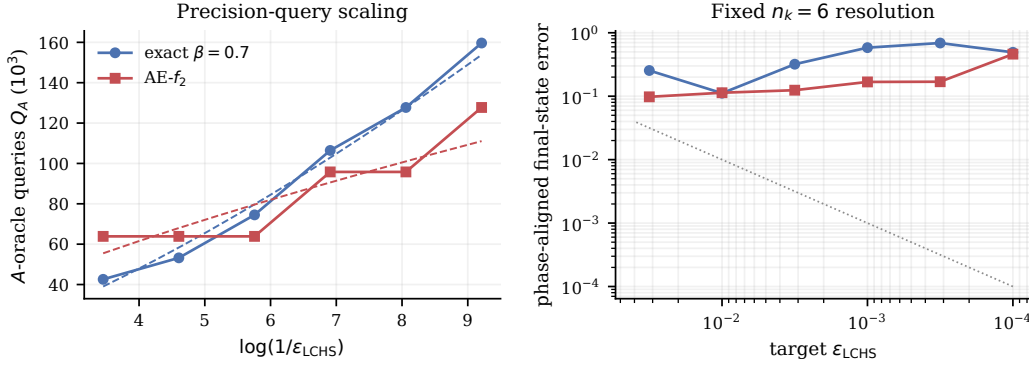


Figure 1: **Left:** expanded A-oracle queries for all six precision targets; dashed curves are the predeclared fits in Eq. (10). **Right:** independent phase-aligned state errors extracted from the same QuCF outputs. The dotted diagonal is a target guide, not a fit. The mismatch exposes fixed- n_k quadrature saturation, so we do not relabel the contour target as achieved end-to-end error.

Emulation environment. Circuits were parsed and fully expanded by the supplied GPU QuCF implementation on one NVIDIA H200 (compute capability 9.0, driver 580.95.05), backed by QuEST [13]. The source was built without modifying the emulator; the kernel generator emits ordinary QuCF oracle and QSVT files. The binary, inputs, outputs, logs, and environment have SHA-256 records. STMC is QuCF’s expanded gate-set count. Every final metric below is read from an executed HDF5 output, not estimated from the continuous LCHS sum.

5 Results

5.1 Primary comparison

Table 2 reports the headline pair and every predeclared characterization pair. At the headline, AE- f_2 lowers Q_A by 20.0% despite its three selector invocations. It also has lower error (0.4604 versus 0.4917) and higher success (0.691 versus 0.209), satisfying both parity conditions. Ancilla counts are equal because the substitution uses the same registers. The STMC column charges the state-level Grover iterate, both reflections, and every weight operation; its finite-point behavior reflects the competition between that overhead and the smaller selector cutoff. At the headline the cutoff reduction also lowers STMC, but this is not uniform over the loose-target regime. Gate complexity was not the predeclared scaling endpoint and is not selectively filtered below.

Figure 1 shows all six executed precision points. The exact kernel gives $p = 1.399 \pm 0.093$ (95% CI $[1.216, 1.581]$, $R^2 = 0.983$), consistent with $1/0.7 = 1.429$. AE- f_2 gives $p = 0.708 \pm 0.192$ (CI $[0.332, 1.083]$, $R^2 = 0.773$), consistent with the predicted $p = 1$. Their difference is 0.691 ± 0.213 ; the normal test gives $z = 3.24$, directional $p = 5.93 \times 10^{-4}$, and two-sided $p = 1.19 \times 10^{-3}$. The stair steps are not noise: K must be a multiple of four, and Eq. (7) charges complete QSP segments.

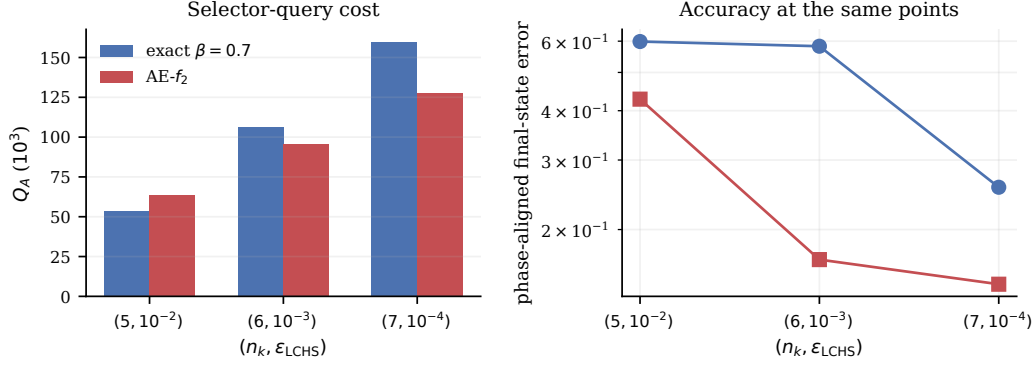


Figure 2: Complete predeclared family characterization. Query cost (left) is paired with the independently emulated final-state error (right). At the coarse point the cheaper exact kernel and more accurate analytic kernel form a tradeoff; the latter dominates at the two tighter points.

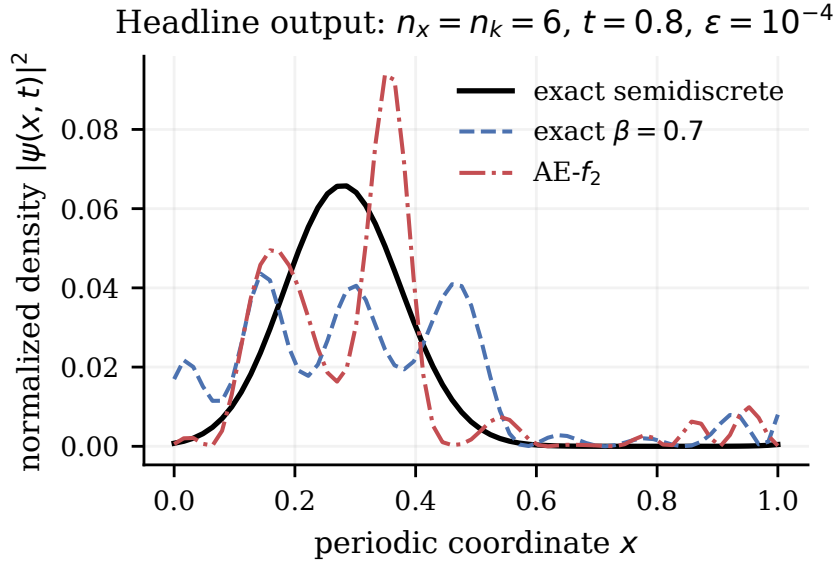


Figure 3: All-zero-ancilla headline output at $\epsilon_{\text{LCHS}} = 10^{-4}$. Each circuit state is normalized only after its success probability is recorded.

173 5.2 Which kernel wins at finite resolution?

174 The cross-resolution result is a crossover rather than uniform dominance (Fig. 2). At $(n_k, \epsilon_{\text{LCHS}}) =$
 175 $(5, 10^{-2})$, AE- f_2 costs 1.20 times the exact-kernel queries: the reduction from $K = 20$ to $K = 8$ is
 176 insufficient to repay three selector calls. The exact kernel is therefore cheaper at this coarse target,
 177 while the variant is more accurate (the error changes from 0.5988 to 0.4279) and has higher success;
 178 neither family dominates that point. At $(6, 10^{-3})$, the ratio falls to 0.90 and the errors are 0.5826
 179 versus 0.1679. At $(7, 10^{-4})$, it falls further to 0.80, with errors 0.2560 and 0.1454. Hence AE-
 180 f_2 dominates in the resolved, tighter-target regime. At coarse targets, g_β remains preferable only
 181 when selector calls, rather than achieved accuracy or success, set the decision. This mechanism
 182 also explains why the analytic family is asymptotically optimal in this implementation: its bounded
 183 normalization cost is constant, while the cutoff advantage grows with precision.

184 Figure 3 visualizes the normalized headline density. The plot is not used to compute Eq. (9); both
 185 curves and that metric are independently extracted from the complex state vector. The analytic
 186 extension tracks the runtime-constructed semidiscrete reference more closely.

5.3 Failure modes and classical compilation cost

The experiment exposes three costs that asymptotic notation hides. First, AE- f_2 loses in queries at the two loosest $n_k = 6$ targets and at the coarse $n_k = 5$ characterization point because of the fixed success-restoring Grover iterate. Second, at $n_k = 6$ the state error does not follow the decreasing contour target monotonically. Increasing K spreads the same 64 FCC nodes over a wider interval, and the nonlinear kernel–evolution product becomes quadrature-limited. The certified ϵ_{LCHS} controls only the continuous-kernel tail plus strip error in Eq. (4); total error additionally contains FCC, selector, and QSVT contributions. Third, an exact controlled phase table makes kernel phases reliable but contributes exponentially many multi-controlled rotations in n_k . It is benign for the chosen query metric, not for large-register gate counts.

A pre-analysis circuit audit also rejected an implementation path that placed the inverse of the right complex preparation on the selector’s left. That path produces $|w_j|$ by canceling the phase. Fourteen completed outputs were discarded, the conjugate left preparation described above was inserted, and all sixteen circuits were rerun; none of the rejected states enters a table, fit, or figure. The retained-input parser assertion and the independent finite-sum comparison guard against this otherwise silent failure.

Classically, the six multi-start contour optimizations and 50-decimal certifications took 94.6 seconds. Synthesis of all sixteen QSVT angle sets took 0.195 seconds in total (0.009–0.024 seconds each). Their standalone emulations expanded to 362–1,652 STMC gates, and every one passed before a full circuit was generated. The 16 end-to-end emulator executions consumed 1.01 aggregate GPU-simulation hours (the rejected phase path is excluded from this retained-run total). These values show that contour optimization dominates kernel-side classical work at this scale; full state-vector evolution dominates the verification cost.

6 Discussion and limitations

The experiment cleanly transfers the optimal analytic-extension mathematics into an explicit selector that was designed for exact kernels. The measured exponent is lower, the theoretical exponents lie inside their respective intervals, and the most stringent matched configuration improves queries, state error, and success simultaneously. The finite-resolution crossover provides an actionable rule: for loose targets, one exact-kernel selector call minimizes target-conditioned queries but does not dominate both accuracy and success; use AE- f_2 once the exact-kernel cutoff exceeds roughly three times the rounded analytic cutoff, where it dominates all three outcomes. This is a circuit-dependent statement, not a claim that f_2 is universally best.

Several limitations delimit the conclusion. We study one time-independent ADE, one spatial resolution, one time, and only $n_k \leq 7$; KvN dynamics and time-dependent generators remain untested. The fitted range is 2.5 decades and contains cutoff quantization, so the confidence intervals are more informative than the point estimates. The final amplitude amplification is state-level and fixed from a pre-validation normalization analysis; other inputs can require a different constant. STMC is an emulator gate metric, not a fault-tolerant T count. The basis-controlled phase table is exact but does not scale favorably in gates. Finally, the f_2 parameters minimize the published continuous cost $\alpha_R R$, not total FCC error or total STMC; joint optimization could move the practical crossover, but would no longer isolate the kernel substitution. The more general $f_{i,j}$ family can improve the theoretical constant [16], and analytic QSP angle constructions may reduce classical synthesis [3]; both are natural next experiments.

7 Conclusion

An optimal-class analytic-extension kernel can be compiled into the economical FCC–single-QSP LCHS framework without changing its selector, quadrature, qubitization, or explicit dissipative block encoding. Actual QuCF executions show a statistically lower precision exponent and a 20.0% headline query reduction at better state accuracy and success probability. Equally importantly, the full sweep shows where substitution is not a strict improvement: at coarse precision fixed amplification produces a tradeoff—the exact kernel is cheaper at several points but does not dominate accuracy and success. Separating a continuous kernel certificate, a discrete weight circuit, and an executed dynamics circuit turns the predicted optimal scaling into a reproducible end-to-end result.

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A Complete end-to-end sweep

Table 3 contains every distinct predeclared full-circuit run, including precision points omitted from the compact primary table. d_w is the selected QSVT polynomial degree and K is the selector cutoff.

Table 3: All full QuCF executions; no sweep point is filtered.

Method	n_k	ϵ_{LCHS}	K	d_w	Final error	Success	STMC	Q_A
$\beta = 0.7$ exact	5	1.0×10^{-2}	20	32	0.59877	0.2889	2,585,205	53,220
AE- f_2	5	1.0×10^{-2}	8	32	0.42785	0.8831	3,109,470	63,864
$\beta = 0.7$ exact	6	3.2×10^{-2}	16	62	0.25425	0.2283	2,115,492	42,576
AE- f_2	6	3.2×10^{-2}	8	62	0.09776	0.8808	3,185,184	63,864
$\beta = 0.7$ exact	6	1.0×10^{-2}	20	62	0.11152	0.2155	2,645,415	53,220
AE- f_2	6	1.0×10^{-2}	8	62	0.11336	0.7880	3,185,184	63,864
$\beta = 0.7$ exact	6	3.2×10^{-3}	28	62	0.31931	0.2363	3,699,181	74,508
AE- f_2	6	3.2×10^{-3}	8	62	0.12436	0.7191	3,185,184	63,864
$\beta = 0.7$ exact	6	1.0×10^{-3}	40	62	0.58258	0.2741	5,279,830	106,440
AE- f_2	6	1.0×10^{-3}	12	62	0.16785	0.6681	4,774,953	95,796
$\beta = 0.7$ exact	6	3.2×10^{-4}	48	80	0.68908	0.2992	6,336,368	127,728
AE- f_2	6	3.2×10^{-4}	12	62	0.16911	0.6227	4,774,953	95,796
$\beta = 0.7$ exact	6	1.0×10^{-4}	60	80	0.49168	0.2094	7,920,849	159,660
AE- f_2	6	1.0×10^{-4}	16	62	0.46044	0.6907	6,355,602	127,728
$\beta = 0.7$ exact	7	1.0×10^{-4}	60	126	0.25596	0.2152	8,099,675	159,660
AE- f_2	7	1.0×10^{-4}	16	126	0.14540	0.5854	6,520,920	127,728

B Standalone complex-weight validation

Table 4 reports the actual isolated QSVT/SIN/phase circuits. The amplitude error is phase-aligned ℓ_∞ error against Eq. (6); the probability error compares the measured and predicted all-zero-signal probabilities before weight amplification. The full degree-attempt sequences, guard maxima, synthesized angles, target arrays, and HDF5 outputs are retained in the result bundle.

Table 4: All standalone kernel-weight circuits. r_w is the number of weight-preparation amplification rounds.

Method	n_k	ϵ_{LCHS}	d_w	r_w	STMC	max amplitude error	probability error
$\beta = 0.7$ exact	5	1.0×10^{-2}	32	2	362	4.83e-14	1.54e-14
AE- f_2	5	1.0×10^{-2}	32	2	362	4.89e-14	1.73e-14
$\beta = 0.7$ exact	6	3.2×10^{-2}	62	2	757	3.96e-14	1.41e-14
AE- f_2	6	3.2×10^{-2}	62	2	757	8.86e-14	1.90e-14
$\beta = 0.7$ exact	6	1.0×10^{-2}	62	3	757	7.43e-14	3.24e-14
AE- f_2	6	1.0×10^{-2}	62	2	757	1.08e-13	2.13e-14
$\beta = 0.7$ exact	6	3.2×10^{-3}	62	3	757	2.94e-14	7.90e-15
AE- f_2	6	3.2×10^{-3}	62	2	757	1.28e-13	2.35e-14
$\beta = 0.7$ exact	6	1.0×10^{-3}	62	3	757	2.49e-14	4.70e-15
AE- f_2	6	1.0×10^{-3}	62	3	757	7.15e-14	8.74e-16
$\beta = 0.7$ exact	6	3.2×10^{-4}	80	3	955	3.88e-14	6.50e-15
AE- f_2	6	3.2×10^{-4}	62	3	757	9.83e-14	8.27e-15
$\beta = 0.7$ exact	6	1.0×10^{-4}	80	4	955	4.13e-14	2.92e-14
AE- f_2	6	1.0×10^{-4}	62	3	757	2.19e-14	5.02e-15
$\beta = 0.7$ exact	7	1.0×10^{-4}	126	5	1,652	1.23e-14	5.83e-16
AE- f_2	7	1.0×10^{-4}	126	3	1,652	7.68e-15	6.94e-18

297 C Circuit register and cost audit

298 For $n_x = n_k = 6$, the executed circuit has 24 qubits: six state qubits, six FCC qubits, two initial-
299 state ancillae, one weight-amplification flag (reused by the final reflections), two weight-signal ancil-
300 lae, two selector ancillae, and five ADE block-encoding ancillae. The register declaration is identical
301 between kernels. QuCF expands inverse circuits and repeats before writing `basic/N_gates`; this
302 HDF5 value is the reported STMC count. It also writes the selector repeat `nt`; together with the ac-
303 tual selector-angle dataset size, the two-block-encoding-per-walk expansion in the supplied source,
304 and the number of complete selector invocations parsed from the executed input, this reconstructs
305 Eq. (7). The analysis aborts if reconstructed and predeclared query counts differ.

306 The ADE block encoding explicitly implements both centered-advection shifts and diffusion shifts.
307 Its rotations are algebraically regenerated for each K from $v/[D(2^{n_x} - 1)]$ and reproduce the sup-
308 plied $K = 20, 40, 80$ constants. Hence a query counted above reaches an actual block-encoding
309 circuit; it is not an external black box omitted from STMC.

310 D Reproducibility and traceability

311 The frozen scientific plan is in `code/kernel_plan.json` with SHA-256 prefix `38acf28c0068`; the
312 complete digest is recorded in every downstream manifest. `code/kernel_derivation/select_`
313 `parameters.py` creates the certified contour log; `generate_weight_angles.py` and `validate_`
314 `weight_circuits.py` create and execute the isolated QSVT circuits; `generate_full_runs.py`
315 emits the selectable full circuits; and `run_full_sweep.py` executes them. `analyze_results.py`
316 refuses an incomplete manifest, reconstructs the exact ADE reference, reads every metric, and gen-
317 erates all table rows and numerical macros in this paper. `make_figures.py` similarly requires all
318 sixteen outputs and records every plotted point plus source hashes in `figures/figure_data.json`.
319 Raw emulator inputs, stdout, circuit logs, and HDF5 outputs reside under `logs/full_runs/`;
320 standalone records reside under `logs/weight_angles/`. `attempts_log.md` discloses rejected
321 kernels, resource objectives, pre-validation corrections, build failures, and the discarded phase-
322 canceling circuit path, while `codebase_notes.md` records the boundary between supplied and new
323 code.